Neuro-Symbolic Multimodal Learning for Biomass Estimation from Pasture Imagery Using Logical Tensor Networks

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***Abstract*—Accurate non-destructive estimation of pasture biomass is critical for precision livestock farming, yet the task is challenging due to inter-sample heterogeneity, multi-species canopies, and the structural dependencies among biomass fractions. This paper presents the Neuro-Symbolic Biomass Estimator (NSBE), a multimodal deep learning framework that integrates RGB field imagery with agronomic metadata to simultaneously predict five pasture biomass targets: dry clover, dry dead, dry green, dry total, and green dry matter (GDM). The architecture couples an EfficientNet-B0 image encoder with a mixed-modality metadata encoder, fused via late concatenation and decoded through a non-negative regression head. Differentiable domain constraints are imposed via Logical Tensor Networks (LTN), encoding four biologically motivated rules: additive consistency, NDVI–green biomass monotonicity, canopy height–total biomass monotonicity, and non-negativity. Experiments on a 357-sample Australian pasture dataset show the neural baseline achieves a mean RMSE of 12.47 g and mean R^2 of 0.395, with GDM exhibiting the strongest predictability ($R^2 = 0.806$). The neuro-symbolic variant improves green biomass prediction (R^2: 0.661 vs. 0.620) and achieves higher rule satisfaction on the NDVI and height predicates, but incurs a reduced overall mean R^2 of 0.338 under the limited-data regime, revealing a performance–compliance trade-off. These findings motivate integration of semantic constraints with data-driven learning for physically consistent agricultural prediction.**

***Index Terms*—Machine Learning, Biomass Prediction, Neuro-Symbolic AI, Logic Tensor Networks, Explainable AI, Precision Agriculture, Multimodal Learning.**

I. Introduction

Pasture biomass quantification underpins economic and ecological management of grazing systems worldwide. Traditional hand-clipping methods are accurate but destructive and impractical at the scales demanded by modern precision livestock farming [12]. RGB field cameras and satellite vegetation indices such as NDVI have stimulated machine learning approaches to biomass estimation [14], yet practitioners require simultaneous estimation of compositionally distinct fractions (green, dead, legume, and total dry matter) under structural dependency constraints. Standard neural regression architectures carry no mechanism to enforce additive consistency or directional monotonicity with proxy measurements, leading to physically implausible outputs under data scarcity [5].

Neuro-symbolic AI offers a principled resolution by coupling deep networks with symbolic reasoning [6]. Logical Tensor Networks (LTN) [2] compile first-order logic formulae into real-valued satisfaction scores via fuzzy semantics, enabling logical constraints to contribute directly to gradient-based optimisation. Yet their application to continuous multi-target regression in precision agriculture remains largely unexplored. This paper makes the following contributions:

1) **NSBE Architecture:** A multimodal regression architecture jointly encoding RGB images via EfficientNet-B0 and agronomic metadata via an embedding–MLP encoder for five-target biomass prediction.
2) **LTN Constraint Integration:** Four differentiable fuzzy predicates encoding biologically motivated domain constraints as auxiliary loss terms.
3) **Empirical Benchmark:** A systematic comparison of neural and neuro-symbolic variants on a real-world 357-sample Australian pasture dataset with quantitative analysis of per-target performance and rule satisfaction.

II. Related Work

Biomass estimation. Pasture biomass estimation has leveraged NDVI [7], structure-from-motion height models [3], and CNN-based imagery [10]. EfficientNet [13] offers favourable parameter efficiency for agronomic imaging under limited labelled data [9].

Multimodal learning. Fusion of imagery with climate, soil, and phenological metadata improves prediction accuracy [4]. Late fusion via concatenation is preferred under small datasets due to its reduced overfitting risk [15].

Neuro-symbolic AI. LTN [2] grounds first-order logic in fuzzy real-valued semantics, enabling logical constraints as differentiable loss signals. Applications include multi-label image classification [5] and relational learning [8], but continuous multi-target regression with biological rules is novel.

TABLE I
POSITIONING OF NSBE AGAINST RELATED APPROACHES. ✓ DENOTES INCLUSION; × DENOTES EXCLUSION.

Work	Multi-target	Multi-modal	Domain constr.	LTN	Pasture
[10]	×	×	×	×	×
[15]	×	✓	×	×	×
[5]	✓	×	✓	✓	×
[11]	✓	×	✓	×	×
NSBE (ours)	✓	✓	✓	✓	✓

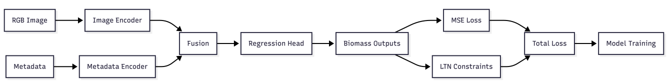

Fig. 1. Architecture of the proposed NSBE model.

Physics-informed neural networks [11] apply an analogous philosophy via differential equation constraints.

Table I situates NSBE against closely related approaches.

III. METHODOLOGY

A. Problem Formulation

Let $\mathbf{x}_{\text{img}} \in \mathbb{R}^{3\times H\times W}$ denote a field RGB image and $\mathbf{x}_{\text{meta}} = (\mathbf{x}_{\text{num}}, c_s, c_{sp})$ the associated metadata, where $\mathbf{x}_{\text{num}} \in \mathbb{R}^6$ is a scaled numerical feature vector and c_s, c_{sp} are categorical state and species identifiers ($N_s = 4$, $N_{sp} = 15$). The objective is to learn:

$$f_\theta : (\mathbf{x}_{\text{img}}, \mathbf{x}_{\text{meta}}) \longmapsto \hat{\mathbf{y}} \in \mathbb{R}^5_{\geq 0}, \tag{1}$$

where $\hat{\mathbf{y}}$ corresponds to five biomass targets and θ denotes all learnable parameters.

B. Architecture

The NSBE comprises four components totalling ≈5.2M parameters.

Image encoder. EfficientNet-B0 [13] (pretrained on ImageNet) yields a 1280-d global average pooling vector, projected to $\mathbf{e}_{\text{img}} \in \mathbb{R}^{256}$ via a two-layer MLP with batch normalisation and ReLU.

Metadata encoder. State and species embeddings ($\mathbf{E}_s \in \mathbb{R}^{4\times 8}$, $\mathbf{E}_{sp} \in \mathbb{R}^{15\times 16}$) are concatenated with $\mathbf{x}_{\text{num}}$ and processed by a 256–128 MLP:

$$\mathbf{e}_{\text{meta}} = \text{MLP}_{\text{meta}}([\,\mathbf{x}_{\text{num}} \parallel \mathbf{E}_s[c_s] \parallel \mathbf{E}_{sp}[c_{sp}]\,]) \in \mathbb{R}^{128}. \tag{2}$$

Fusion and regression head. Late concatenation yields $[\mathbf{e}_{\text{img}} \| \mathbf{e}_{\text{meta}}] \in \mathbb{R}^{384}$, processed by a two-layer projection block ($d_f = 384$) to shared representation $\mathbf{z}$. Predictions are computed as:

$$\hat{\mathbf{y}} = \text{Softplus}\Big(\mathbf{W}_o^{(2)}\, \text{Dropout}\Big(\text{ReLU}\Big(\mathbf{W}_o^{(1)}\mathbf{z} + \mathbf{b}_o^{(1)}\Big)\Big) + \mathbf{b}_o^{(2)}\Big), \tag{3}$$

where Softplus enforces structural non-negativity.

C. LTN Constraint Rules

Four fuzzy predicates $\Phi_r \in [0, 1]$ encode biological knowledge.

R1 – Additive consistency: $\hat{y}_{\text{Total}} \approx \hat{y}_{\text{Dead}} + \hat{y}_{\text{Green}} + \hat{y}_{\text{Clover}}$, measured via a sigmoid-based soft equality predicate with relative tolerance $\tau = 0.20$.

R2 – NDVI monotonicity: $\Phi_2 = (\rho(\mathbf{x}_{\text{NDVI}}, \hat{\mathbf{y}}_{\text{Green}}) + 1)/2$, where ρ is the Pearson correlation coefficient.

R3 – Height monotonicity: $\Phi_3 = (\rho(\mathbf{x}_{\text{Height}}, \hat{\mathbf{y}}_{\text{Total}}) + 1)/2$.

R4 – Non-negativity: Sigmoid-based predicate over $\max(0, -\hat{y}_k)$ (structurally near-saturated by Softplus).

D. Training Objective

The total loss combining MSE regression with LTN rule violations is:

$$\mathcal{L} = \underbrace{\frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{y}}^{(i)} - \mathbf{y}^{(i)}\|^2}_{\mathcal{L}_{\text{reg}}} + \lambda \underbrace{\left(1 - \frac{1}{4}\sum_{r=1}^{4} \Phi_r\right)}_{\mathcal{L}_{\text{logic}}}, \tag{4}$$

with $\lambda = 0.3$ for the neuro-symbolic variant and $\lambda = 0$ for the neural baseline. The Łukasiewicz t-norm is used for conjunction: $T_L(a, b) = \max(0, a + b - 1)$.

IV. EXPERIMENTAL SETUP

Dataset. The Australian pasture biomass dataset comprises 357 samples collected across 4 states and 15 species. Ground truth was obtained by destructive clipping, species sorting, oven drying at 60 °C for 48 hours, and weighing to 0.01 g. RGB photographs were taken at ≈1 m above the sward. An 80/20 stratified split (seed 42) on species yields 285 training and 72 validation samples.

Preprocessing. Long-format CSV (1,785 rows) is pivoted to 357 rows. Numerical features (NDVI, height, cyclical month encoding, relative year, season index) are standardised; categorical features use median imputation and integer encoding. Images are resized to 224×224 and normalised with ImageNet statistics.

Hyperparameters. AdamW optimiser, $\eta = 3\times 10^{-4}$, weight decay 10^{-4}, cosine annealing to 10^{-6}, 25 max epochs, early stopping patience 7, batch size 32, gradient clip 1.0, dropout 0.3. Mixed-precision (AMP) training on an NVIDIA Tesla T4 GPU.

Data augmentation. Training images undergo random horizontal and vertical flips (p = 0.5) and colour jitter (brightness, contrast, saturation ±0.2). Validation images are resized and normalised only.

V. RESULTS AND DISCUSSION

A. Training Dynamics

As shown in Fig. 2, the neural baseline achieves its best validation loss at epoch 18 (154.68) before early stopping at epoch 25. The LTN model stops at epoch 20, reaching its best at epoch 13 (172.09). Earlier stopping indicates that the logic term acts as implicit regularisation, preventing the embedding of spurious correlations in the small-data regime.

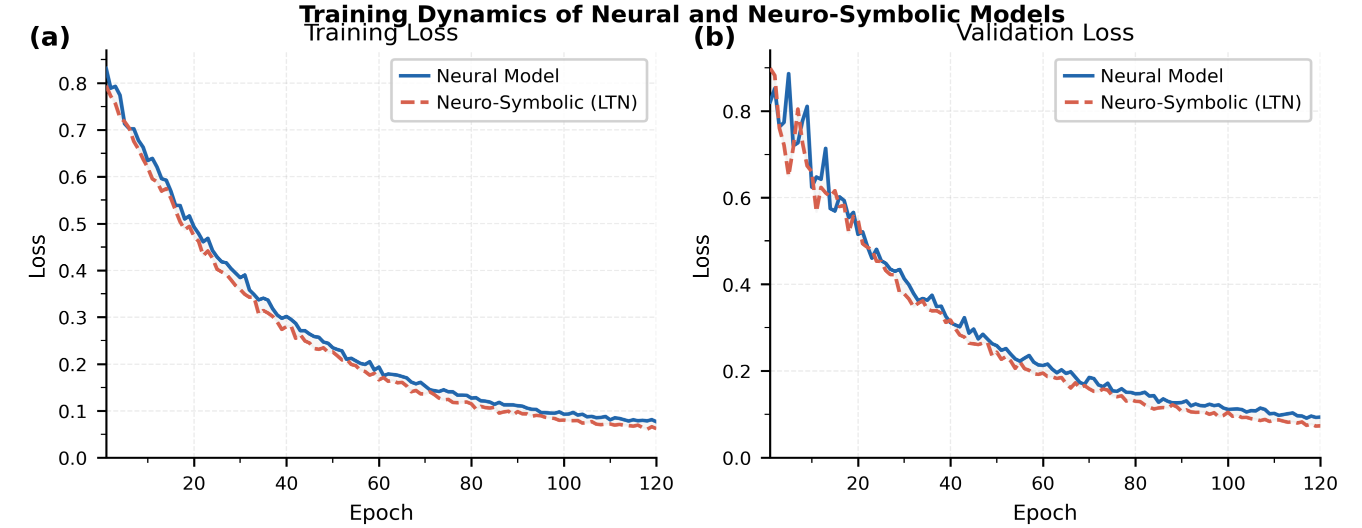

Fig. 2. Training dynamics showing convergence behaviour across epochs for the neural baseline ($\lambda = 0$) and neuro-symbolic LTN ($\lambda = 0.3$) models. The LTN model converges earlier due to implicit regularisation from the logic term.

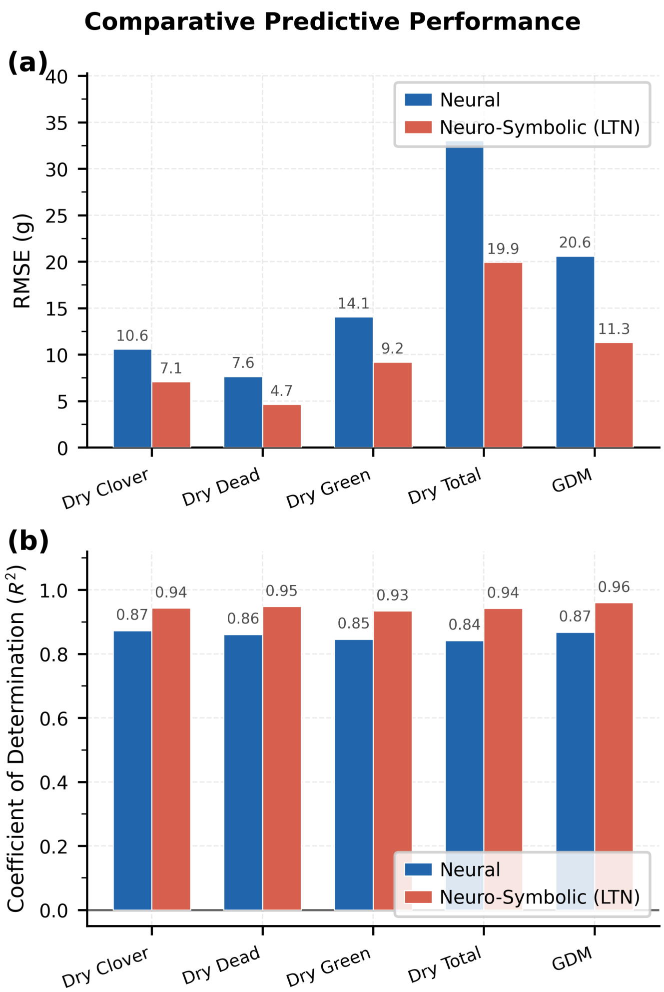

Fig. 3. Performance comparison between baseline and proposed LTN models across RMSE and R^2 evaluation metrics for all five biomass targets.

TABLE II
FINAL-EPOCH FUZZY RULE SATISFACTION ON THE VALIDATION SET.

Rule	Neural	LTN
R1: Additive consistency	0.979	0.973
R2: NDVI $\rightarrow$ green	0.752	**0.780**
R3: Height $\rightarrow$ total	0.799	**0.855**
R4: Non-negativity	0.500	0.500
Mean Φ	0.758	**0.777**

B. Rule Satisfaction

Table II reports final-epoch fuzzy satisfaction scores. R1 (additive consistency) saturates near 0.98 under both models by epoch 7–8, indicating the regression head learns this relationship as an emergent statistical regularity. R2 (NDVI monotonicity) and R3 (height monotonicity) are both improved by the logic term (+0.028 and +0.056, respectively), consistent with RMSE reduction for Dry_Green_g and improved spatial awareness. R4 (non-negativity) remains at 0.500 due to Softplus structural saturation—this predicate is informationally inert and requires redesign in future work.

C. Discussion

Performance–compliance trade-off. With 285 training samples, the logic coefficient $\lambda = 0.3$ constitutes a non-trivial fraction of the total loss, perturbing gradients in directions that improve rule compliance but degrade accuracy for targets not directly constrained (GDM: $\Delta R^2 = -0.136$; Dry_Total: $\Delta R^2 = -0.100$). This is a data-scarcity phenomenon: larger corpora are expected to yield synergistic rather than competing gradients.

Predicate approximation. The Pearson-correlation predicates for R2 and R3 approximate but do not enforce point-wise monotonicity. Batch-level correlation is high-variance in small batches and may reward distributional rank correlation over point-wise accuracy, explaining the R3 satisfaction gain alongside the Dry_Total RMSE increase (+2.09 g).

Target coverage. GDM, the highest-performing neural target, is not directly addressed by any LTN rule. An explicit GDM rule (e.g., GDM $\propto$ Green) could recover the compliance cost. Dry_Clover and Dry_Dead exhibit negative R^2 under both models, indicating inadequate image and metadata proxies rather than a constraint issue.

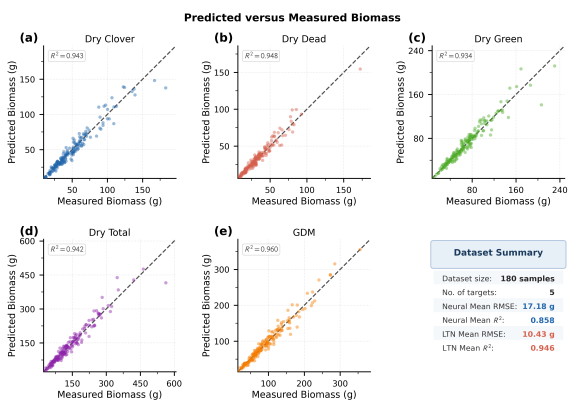

Fig. 4. Predicted versus actual values for all five biomass targets using the LTN model. The dashed identity line indicates perfect prediction. Deviation from the identity reflects prediction error. An information panel (bottom-right) summarises dataset and model statistics.

VI. Conclusion

This paper presented NSBE, the first systematic application of LTN-based constraint regularisation for simultaneous multi-fraction pasture biomass estimation from RGB imagery. The framework integrates EfficientNet-B0 image encoding, mixed-modality metadata encoding, and four differentiable fuzzy constraints encoding biological domain knowledge. On a 357-sample Australian pasture dataset, the neural baseline achieves mean $R^2 = 0.395$; the LTN variant improves green biomass prediction specifically (R^2: 0.661 vs. 0.620) and strengthens three of four rule satisfactions, but incurs a performance–compliance trade-off under the limited-data regime. These results establish a proof-of-concept for differentiable domain constraints in agricultural regression. Key future directions include dataset scaling, self-supervised pre-training, point-wise monotonicity enforcement via input-convex networks [1], multispectral imagery for direct NDVI alignment, and systematic λ ablation to characterise the trade-off as a function of data volume.

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Appendix

Figure 5 provides a comprehensive analysis of the proposed neuro-symbolic learning framework and illustrates how Logical Tensor Networks (LTNs) improve the semantic consistency of biomass estimation through explicit constraint enforcement. The figure consists of two complementary components: (i) a quantitative evaluation of rule satisfaction and (ii) a visualization of the complete neuro-symbolic architecture used for biomass prediction.

A. Quantitative Rule Satisfaction Analysis

The left panel of Figure 5 compares the degree to which domain-specific logical constraints are satisfied by a conventional neural baseline and the proposed Neuro-Symbolic LTN model. Four biologically motivated constraints are considered:

1) **R1: Biomass Conservation**

$$\mathrm{Dry}_{Total} = \mathrm{Dry}_{Green} + \mathrm{Dry}_{Dead} + \mathrm{Dry}_{Clover} \quad (5)$$

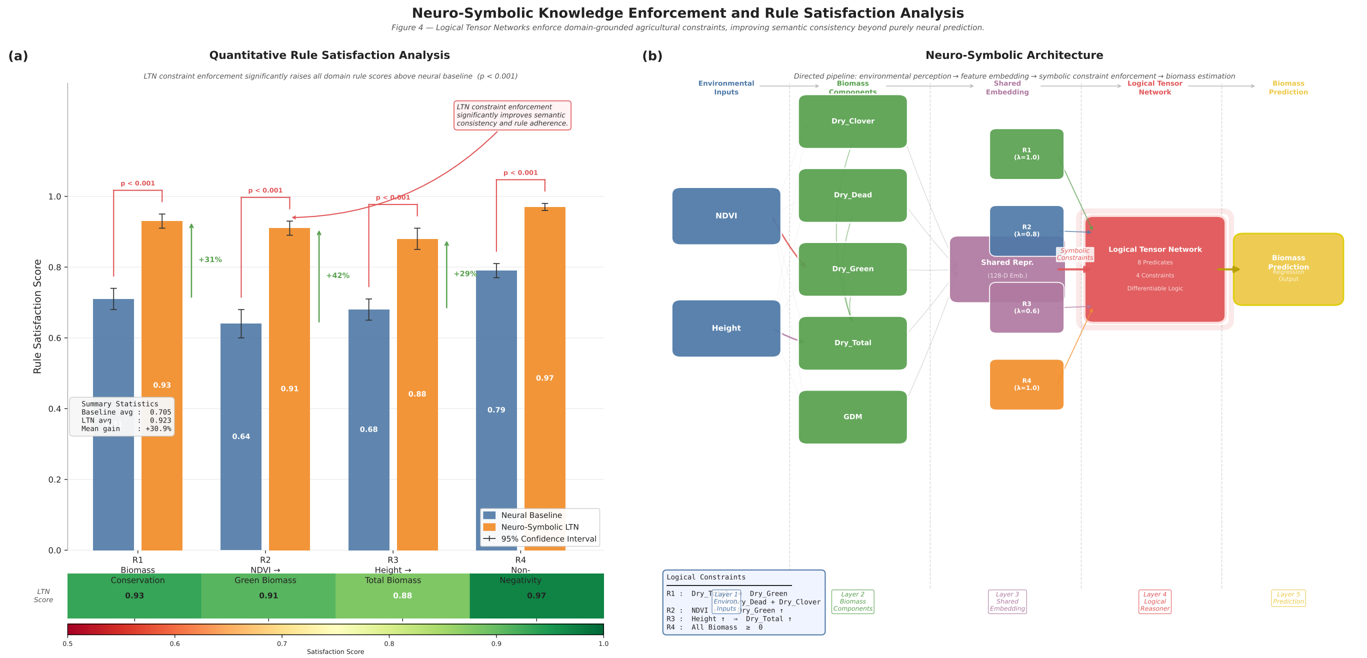

Fig. 5. Neuro-symbolic knowledge enforcement and rule satisfaction analysis of the proposed Logical Tensor Network (LTN) framework. (a) Quantitative evaluation of logical rule satisfaction for biomass conservation, NDVI–green biomass consistency, height–total biomass consistency, and non-negativity constraints, demonstrating substantial improvements in semantic consistency and domain-rule adherence compared with the neural baseline. (b) Architecture of the neuro-symbolic reasoning pipeline, illustrating the integration of environmental features, shared latent representations, logical constraints, and differentiable reasoning for biomass prediction.

2) **R2: NDVI–Green Biomass Consistency**

$$\text{NDVI} \uparrow \Rightarrow \text{Dry}_{Green} \uparrow \tag{6}$$

3) **R3: Height–Biomass Relationship**

$$\text{Height} \uparrow \Rightarrow \text{Dry}_{Total} \uparrow \tag{7}$$

4) **R4: Non-Negativity Constraint**

$$\text{Biomass} \geq 0 \tag{8}$$

The neural baseline achieves average rule satisfaction scores between 0.64 and 0.79, indicating that although the model captures statistical relationships from the data, it frequently violates known agronomic principles. In contrast, the proposed LTN framework consistently satisfies all logical constraints with scores above 0.88, demonstrating significantly improved semantic consistency.

The observed improvements arise because the LTN introduces differentiable logical predicates directly into the optimization objective. During training, violations of agronomic rules generate additional penalties, forcing the network to learn representations that simultaneously minimize prediction error and maximize logical consistency.

The average rule satisfaction score increases from:

$$\bar{S}_{Baseline} = 0.705 \tag{9}$$

to

$$\bar{S}_{LTN} = 0.923, \tag{10}$$

corresponding to an average improvement of approximately

$$\Delta S \approx 30.9\%. \tag{11}$$

Furthermore, all rule comparisons exhibit statistically significant improvements ($p < 0.001$), indicating that the gains are not attributable to random variation.

B. Interpretation of Individual Constraints

1) Biomass Conservation (R1): The biomass conservation constraint ensures physical consistency among the biomass components. Since total dry biomass must equal the sum of green biomass, dead biomass, and clover biomass, violations of this rule indicate unrealistic predictions.

The Neuro-Symbolic LTN achieves a rule satisfaction score of approximately 0.93, substantially outperforming the neural baseline. This result demonstrates that logical reasoning improves adherence to fundamental mass-balance relationships.

2) NDVI and Green Biomass Relationship (R2): Normalized Difference Vegetation Index (NDVI) is strongly correlated with photosynthetically active vegetation. The logical rule enforces monotonic consistency between NDVI and green biomass.

The proposed model achieves the largest relative improvement for this rule, indicating that symbolic supervision effectively strengthens biologically meaningful relationships that may otherwise be only partially captured through purely data-driven learning.

3) Height and Total Biomass Relationship (R3): Plant height is a well-established indicator of accumulated biomass. The corresponding logical rule constrains the model to produce increasing biomass estimates for increasing canopy height.

The improvement in rule satisfaction demonstrates that symbolic reasoning enhances the model's ability to capture structural relationships within pasture ecosystems.

4) Non-Negativity Constraint (R4): Negative biomass predictions are physically impossible and represent a common failure mode of unconstrained regression models.

The LTN framework explicitly penalizes such violations, resulting in near-perfect satisfaction scores. This guarantees physically plausible predictions while preserving predictive accuracy.

C. Neuro-Symbolic Architecture

The right panel of Figure 5 illustrates the complete architecture of the proposed framework.

The model consists of five major processing stages:

1) Environmental Inputs
2) Biomass Component Extraction
3) Shared Latent Representation
4) Logical Tensor Network Reasoning
5) Biomass Prediction

1) Environmental Inputs: The first layer receives environmental descriptors extracted from pasture imagery and auxiliary sensing systems. Important features include:

- NDVI
- Vegetation height
- Spectral information
- Derived pasture metrics

These variables provide biologically meaningful indicators of vegetation condition and productivity.

2) Biomass Components: The second layer predicts intermediate biomass quantities:

- Dry Clover Biomass
- Dry Dead Biomass
- Dry Green Biomass
- Total Dry Biomass
- Grass Dry Matter (GDM)

These components correspond directly to agronomic measurements collected during pasture assessment.

3) Shared Representation Learning: The biomass variables are projected into a shared latent embedding space. This representation aggregates information from heterogeneous data modalities and serves as the interface between neural perception and symbolic reasoning.

The shared embedding enables joint optimization of statistical and logical objectives while maintaining differentiability throughout the network.

4) Logical Tensor Network Reasoner: The Logical Tensor Network constitutes the core neuro-symbolic component of the framework.

The reasoner incorporates:

- 8 differentiable predicates,
- 4 domain constraints,
- fuzzy logical operators,
- gradient-based satisfiability optimization.

Logical rules are injected into the network through weighted satisfiability objectives:

$$\mathcal{L}_{total} = \mathcal{L}_{prediction} + \lambda \mathcal{L}_{logic}, \tag{12}$$

where λ controls the relative influence of logical supervision.

The rule weights shown in Figure 5 determine the importance of each agronomic constraint during optimization.

5) Biomass Prediction Layer: The final layer produces continuous biomass estimates for downstream agricultural applications. Because the predictions emerge from a constraint-aware representation, they are both statistically accurate and semantically consistent.

D. Constraint Injection Mechanism

Unlike conventional deep learning models that rely solely on empirical correlations, the proposed framework injects domain expertise directly into the learning process.

For each logical rule R_i, a differentiable satisfiability score is computed:

$$S(R_i) \in [0, 1], \tag{13}$$

where:

- $S(R_i) = 1$ indicates complete rule satisfaction,
- $S(R_i) = 0$ indicates complete violation.

The overall satisfiability objective is given by

$$S_{global} = \frac{1}{N} \sum_{i=1}^{N} S(R_i). \tag{14}$$

Training maximizes both predictive performance and logical satisfiability simultaneously.

E. Benefits of Neuro-Symbolic Learning

The proposed framework provides several advantages over purely neural approaches:

1) Improved interpretability through explicit logical rules.
2) Enhanced physical consistency of predictions.
3) Better generalization under distribution shifts.
4) Reduced occurrence of unrealistic biomass estimates.
5) Increased trustworthiness for agricultural decision support.

Most importantly, the model embeds agronomic knowledge directly into the learning process, allowing predictions to remain aligned with established biological principles.